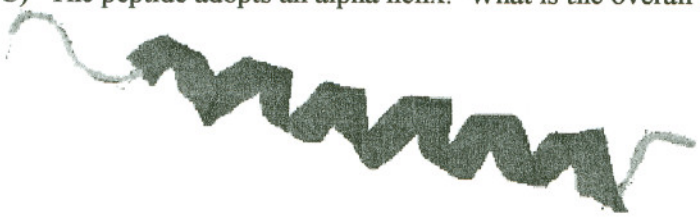
BIOCHEMISTRY CUME
MARCH 7, 2003

1. (60pts) Glucagon has a major role in maintaining normal concentrations of glucose in blood, and is often described as having the opposite effect of insulin. That is, glucagon has the effect of increasing blood glucose levels. Understanding how and why the autonomic nervous system and islet A cell develop defects in glucagon secretion leading to hypoglycemia insensitivity is a major challenge in diabetes research.
Glucagon has the following amino acid sequence:

His-Ser-Gln-Gly-Thr-Phe-Thr-Ser-Asp-Tyr-Ser-Lys-Tyr-Leu-Asp-Ser-Arg-Arg-Ala-Gln-Asp-Phe- Val-Gln-Trp-Leu-Met-Asn-Thr

A) Which of the amino acids in glucagon have non-polar sidechains?
B) Which of the amino acids in glucagons have sidechains that can be a hydrogen bond donor?
C) The peptide adopts an alpha helix. What is the overall length of this helix?



[useful information: 1Å=10^{-1} nm, pitch of alpha helix= 1.5Å per residue, molecular weight per average amino acid = 110 Da]

2. (40 pts) The data bases for protein amino acid sequences now contain several hundred thousand, perhaps more than a million, entries. Biochemists, with nothing better to do, have searched these data bases for cosmic (comic?) messages "C words" of profound meaning spelled out in the single letter amino acid sequence such as ELVIS and LIVES in the same protein. ELVIS has been found several times.

A) What are the pI of these peptides, ELVIS and LIVES?

B) You have a mixture of peptides, ELVIS and LIVES. What type of peptide separation techniques might you use to separate them?

Useful Information:

Side Group		pKa
Arginine	R	12.5
Aspartic acid	D	3.9
Cysteine	C	8.3
Glutamic acid	E	4.1
Histidine	H	6.0
Lysine	K	10.1
Tyrosine	Y	10.1
C-terminus		2.5
N-terminus		9.0